**Corollary .0.1**

Every automorphism of $\mathbb{F}_{p^k}$ is of the form $x \to x^{p^i}$ for some $i \in \mathbb{Z}/k\mathbb{Z}$.

*Proof.* We know that $\mathbb{F}_{p^k}^{\times}$ is a cyclic group. Therefore, the degree of the minimal polynomial $m_\alpha(x)$ of $\alpha$ over $\mathbb{F}_p$ is $[\mathbb{F}_p(\alpha) : \mathbb{F}_p] = k$. Now apply the Frobenius automorphism from $x \to x^p$, then

$$m_\alpha(x)^p = m(\alpha^p) = 0$$

So $\alpha^p$ is a root of $m_\alpha(x)$. Since $x \to x^{p^i}$ is a composition of $x \to x^p$ for $i$ times, we see that $\alpha^{p^i}$ are all roots of $m_\alpha(x)$. Since any automorphism $\sigma$ maps $\alpha$ to other roots in the minimal polynomial, we see that $x \to x^{p_i}$ are all the automorphisms of $\mathbb{F}_{p^k}$. $\mathbb{Z}$

**Corollary .0.2**

$$\operatorname{Aut}(\mathbb{F}_{p^k}) \cong \mathbb{Z}/k\mathbb{Z}$$

**Proposition .0.3**

The map

$$\theta : D \to \operatorname{Aut}(\mathcal{O}_L/\mathfrak{p}_i)$$

defined by $\theta(\sigma) = \overline{\sigma}$ is surjective.

*Proof.* Write $\ell = \mathcal{O}_L/\mathfrak{p}_i$. Then we know that $\ell = \mathbb{F}_p(\overline{\alpha})$ for some $\overline{\alpha} \in \mathcal{O}_L/\mathfrak{p}_i$. We want to show that:

For any $\overline{\alpha}^{p^i}$, there is some $\sigma \in D$ such that $\overline{\sigma}(\overline{\alpha}) = \overline{\alpha}^{p^i}$.

By Chinese Remainder Theorem, we can find $\alpha \in \mathcal{O}_L$ such that

$$\alpha \equiv \overline{\alpha} \pmod{\mathfrak{p}_i}$$
$$\alpha \equiv 0 \pmod{\mathfrak{p}_j} \quad j \neq i$$

Let $f(x)$ be the minimal polynomial of $\alpha$ over $\mathbb{Q}$ and $\overline{g}(x)$ be the minimal polynomial of $\overline{\alpha}$ over $\mathbb{F}_p$. Then $\deg(f) = [\mathbb{Q}(\alpha) : \mathbb{Q}]$ and $\deg(\overline{g}) = [\ell : \mathbb{F}_p]$. The roots of $f(x)$ consisting of those roots that are distinct among $\sigma(\alpha)$. So there is a subset $H \subset G$ such that

$$f(x) = \prod_{\sigma \in H} (x - \sigma(\alpha))$$

Let $H' = H \cap D$. If $\sigma \notin D$, then $\sigma^{-1} \notin D$. Therefore, $\sigma^{-1}(\mathfrak{p}_i) \neq \mathfrak{p}_i$. Since $\alpha \notin \mathfrak{p}_i$, and $\alpha \in \mathfrak{p}_j$ for $j \neq i$, we see that $\alpha \in \sigma^{-1}(\mathfrak{p}_i)$, so $\sigma(\alpha) \in \mathfrak{p}_i$. Reduce $f$ mod $\mathfrak{p}_i$, we get

$$\begin{aligned}\overline{f}(x) &= \prod_{\sigma \in H'} (x - \overline{\sigma(\alpha)}) \prod_{\sigma \in H \setminus H'} (x - \overline{\sigma(\alpha)}) \\ &= x^{|H \setminus H'|} \prod_{\sigma \in H'} (x - \overline{\sigma(\alpha)}).\end{aligned}$$

So all the non-zero roots of $\overline{f}(x)$ are of the form $\overline{\sigma((\alpha))} = \overline{\sigma}(\overline{\alpha})$. In particular, $\overline{f}(\overline{\alpha}) = \overline{f(\alpha)} = 0$. Since $\overline{g}(x)$ is the minimal polynomial of $\overline{\alpha}$, we see that

$$\overline{g}(x) \mid \overline{f}(x)$$

Hence every root of $\overline{g}(x)$ is also a non-zero root of $\overline{f}(x)$, which are those of the form $\overline{\sigma}(\overline{\alpha})$. By the following remark, all the roots of $\overline{g}(x)$ induces all the automorphism of $\ell$. Hence, we complete the proof. $\mathbb{Z}$

**Remark .0.1**

Note that here we use the fact that if we have a Galois extension $L/K$, i.e. the extension is separable and normal, then given $\alpha \in L$ whose minimal polynomial is $f$, and $\alpha'$ another root of $f$, we have an isomorphism $K(\alpha) \cong K(\alpha')$. Moreover, we can extend it into an isomorphism in $\mathrm{Gal}(K/L)$.

**Remark .0.2**

If $\mathcal{O}_L/\mathfrak{p}_i \cong \mathbb{F}_{p^f}$, then

$$\mathrm{Aut}(\mathcal{O}_L/\mathfrak{p}_i) \cong \mathbb{Z}/f\mathbb{Z}$$

So we get a surjective group homomorphism

$$D \twoheadrightarrow \mathbb{Z}/f\mathbb{Z}$$

Let $I$ be the kernel of this map, then

$$f = [\mathcal{O}_L/\mathfrak{p}_i : \mathbb{F}_p] = [D : I]$$

and

$$D/I \cong \mathbb{Z}/f\mathbb{Z}$$

$D/I$ has distinguished generator, called the "Frobenius coset", namely any $\sigma \in D$ that induces the $p$th power map on $\mathcal{O}_L/\mathfrak{p}_i$.

In sum, we get the following useful information:

- $e$ = ramification index of $\mathfrak{p}_i$.
- $f$ = degree of field extension $[\mathcal{O}_L/\mathfrak{p}_\mathfrak{i} : \mathbb{F}_p]$.
- $r = [G : D]$.
- $f = [D : I]$.
- $|G| = efr$
- $|D| = ef$.
- $|I| = e$.
- $p$ unramified in $L/\mathbb{Q} \iff |I| = 1$. In this case, the Frobenius cosets in $D/I$ are just single elements of $D$, called the "Frobenius element" for $\mathfrak{p}_i$.